

PRESCRIBING INFORMATION: For the Use of Registered Veterinary Practitioner or a Hospital or a Laboratory Only.

(Veterinary)

NEOMEC IVERMECTIN INJECTION BP 1% W/V

Composition:

Each ml contains:

Ivermectin BP. 10 mg

Benzyl Alcohol BP 0.015ml

(as preservative)

Propylene Glycol BP qs to 1ml

Target species:

Beef and non-lactating dairy cattle, sheep and pigs

Pharmacodynamics

Ivermectin is a 22,23-dihydro derivative of an avermectin (which is a fermentation product produced by *Streptomyces avermitilis*) and consists of 2 homologues: B1a and B1b. It is a parasiticide with nematocidal, insecticidal and acaricidal activity documented in a wide range of domesticated animals.

Ivermectin is a member of the macrocyclic lactone class of endectocides which have a unique mode of action. Compounds of this class bind selectively and with high affinity to glutamate gated chloride ion channels which occur in invertebrate nerve and muscle cells. This leads to an increase in the permeability of the cell membrane to chloride ions with the hyper-polarisation of the nerve or muscle cell resulting in paralysis and death of the parasite.

Compounds of this class may also interact with other ligand-gated chloride channels, such as those gated by the neurotransmitter gamma-aminobutyric acid (GABA).

The margin of safety for compounds of this class is attributable to the fact that mammals do not have glutamate gated chloride channels, the macrocyclic lactone have a low affinity for other mammalian ligand-gated chloride channels and they do not readily cross the blood-brain barrier.

Pharmacokinetics

In cattle, following subcutaneous administration at the recommended dose of 0.2 mg ivermectin / kg, a maximum concentration of approximately 30.4 ng / mL is observed 3-6 days after injection. A total AUC of the order of 9237 ng.h / mL is calculated

In sheep, following subcutaneous administration at the recommended dose of 0.2 mg ivermectin per kg body weight, peak plasma concentration of approximately 12.5 ng / mL is observed approximately 3-7 days after the injection. A total AUC of the order of 2461 ng.h / mL is calculated.

In swine, following subcutaneous administration at the recommended dose of 0.3 mg ivermectin / kg, a peak concentration of approximately 7 ng / mL is observed 2-4 days after injection. A total AUC of the order of 1122 ng.h / mL is calculated.

Ivermectin is partially metabolized. Only 2% of the dose is eliminated by the urinary tract, the rest being eliminated by the feces. 60% of the product is eliminated in the non-metabolized form in the feces. The remainder is eliminated as metabolites or degradation products.

Biliary excretion and subsequent elimination via feces is certainly the major route of elimination in pigs.

Indication:

Cattle

For the treatment and control of the following species of gastrointestinal roundworms, lungworms, eyeworms, warbles, mites and sucking lice in beef cattle and non-lactating dairy cattle.

Gastrointestinal roundworms (adults and fourth stage larvae):

Ostertagia ostertagi (including inhibited *O. ostertagi*), *Ostertagia lyrata*, *Haemonchus placei*, *Trichostrongylus axei*, *Trichostrongylus colubriformis*, *Cooperia oncophora*, *Cooperia punctata*, *Cooperia pectinata*, *Bunostomum phlebotomum*, *Oesophagostomum radiatum*, *Strongyloides papillosus* (adult), *Nematodirus helvetianus* (adult), *Nematodirus spathiger* (adult), and *Trichuris* spp (adults)

Lungworms (adult and fourth stage larvae):

Dictyocaulus viviparus

Eyeworms (adult):

Thelazia spp

Warbles (parasitic stages):

Hypoderma bovis, *Hypoderma lineatum*

Sucking Lice:

Linognathus vituli, *Haematopinus eurysternus*, *Solenopotes capillatus*

Mange Mites:

Psoroptes bovis, *Sarcoptes scabiei* var *bovis*

The product may also be used as an aid in the control of the biting louse *Damalinia bovis* and the mange mite *Chorioptes bovis*, but complete elimination may not occur.

Sheep

For the treatment and control of psoroptic mange (sheep scab), gastrointestinal nematodes, lungworms and nasal bots of sheep.

At the recommended dosage level of 200 mcg ivermectin per kg bodyweight the product effectively controls the following parasites of sheep:

Gastrointestinal roundworms (adults and fourth stage larvae):

Teladorsagia circumcincta, O. trifurcata, Haemonchus contortus, Trichostrongylus axei (adults), Trichostrongylus colubriformis, Trichostrongylus vitrinus (adults), Cooperia curticei, Oesophagostomum venulosum, Oesophagostomum columbianum, Nematodirus filicollis, Chabertia ovina, Trichuris ovis (adults)

Inhibited larval stages and benzimidazole resistant strains of Haemonchus contortus and Teladorsagia circumcincta are also controlled.

Lungworms:

Dictyocaulus filaria (adults and fourth stage larvae), Protostrongylus rufescens (adults).

Mange Mites:

Psoroptes ovis

Nasal Bot:

Oestrus ovis (all larval stages)

Pigs

For the treatment and control of the harmful species of gastrointestinal roundworms, lungworms, lice and mange mites of pigs. At the recommended dose rate of 300ug/kg the product provides effective control of the following parasites:

Gastrointestinal worms:

Ascaris suum (adults and fourth-stage larvae)

Hyostrongylus rubidus (adults and fourth-stage larvae)

Oesophagostomum spp (adults and fourth-stage larvae)

Strongyloides ransomi (adults and somatic larval stages)

Lungworms:

Metastrongylus spp (adults)

Lice:

Haematopinus suis

Mange Mites:

Sarcoptes scabiei var suis

The product may also be used as an aid in the control of adult whipworm (Trichuris suis).

Dosage and Administration

To ensure administration of a correct dose, body weight should be determined as accurately as possible; accuracy of the dosing device should be checked.

Cattle

Ivermectin should be administered at a dosage rate of 200 µg per kg bodyweight (1 ml to 50 kg bodyweight). It should only be injected subcutaneously in front of or behind the shoulder using aseptic technique. A sterile 17 gauge x ½ inch needle is recommended.

Sheep

Administer only by subcutaneous injection in the neck at the recommended dosage level of 200 µg ivermectin per kg bodyweight using aseptic technique. Each ml contains 10 mg ivermectin to treat 50 kg of bodyweight. Swab the septum before removing each dose. Use a dry sterile needle and syringe.

Administer subcutaneously only. Inject once under the loose skin in the neck. For the treatment and control of sheep scab (*Psoroptes ovis*) two injections with a seven day interval are required to treat clinical signs of scab and eliminate living mites. Use of a 17 gauge x ½ inch (15-20 mm) needle is suggested. Replace with a fresh sterile needle after every 10-12 animals. Injection of wet or dirty animals is not recommended.

When treating sheep of less than 16 kg seek veterinary advice regarding the use of 1 ml disposable syringes graduated in increments of 0.1 ml.

For the treatment of individual sheep a syringe not exceeding 2.0 ml and calibrated in increments of 0.1 ml should be used.

Pigs

Administer at a dosage rate of 300 µg per kg bodyweight (1 ml per 33 kg). The product should be injected subcutaneously into the neck using aseptic technique. A sterile 17 gauge x ½ inch needle is recommended

Contraindications

The product is for administration only by the subcutaneous route and must not be given via other routes. It should not be used in other species than those indicated as severe reactions, including fatalities in dogs, may occur.



Precautions for Use:**Special Warnings for Each Target Species:**

Immediately following subcutaneous injection, activity suggesting pain, sometimes intense but usually transient, has been observed in some sheep

Special Precautions for Use:

Care should be taken to avoid the following practices because they increase the risk of development of resistance and could ultimately result ineffective therapy:

- Too frequent and repeated use of anthelmintics from the same class, over an extended period of time.
- Underdosing, which may be due to underestimation of body weight, misadministration of the product, or lack of calibration of the dosing device (if any).

Suspected clinical cases of resistance to anthelmintics should be further investigated using appropriate tests (e.g. Faecal Egg Count Reduction Test). Where the results of the test(s) strongly suggest resistance to a particular anthelmintic, an anthelmintic belonging to another pharmacological class and having a different mode of action should be used.

Resistance to ivermectin (an avermectin) has been reported in *Teladorsagia* in sheep and goats within the EU and it is common in *Haemonchus* in sheep outside the EU. It has been reported in *Cooperia oncophora* in cattle within the EU, in *Teladorsagia* in cattle in developed countries such as New Zealand and *Haemonchus* outside the EU. Therefore the use of this product should be based on local (regional, farm) epidemiological information about the susceptibility of nematodes and recommendations on how to limit further selection for resistance to anthelmintics.

(i) Special precautions for use in animals

In pigs, especially those under 16 kg for which less than a volume of 0.5 ml is indicated, dosing accuracy is important. The use of a suitably calibrated syringe that can accurately deliver 0.1 ml is recommended.

(ii) Special precautions to be taken by the person administering the veterinary medicinal product to animals

Do not eat, drink or smoke while handling the product.

Direct contact of the product with the skin should be kept to a minimum. Wash hands after use.

Take care to avoid accidental self-injection: this product may cause local irritation and/or pain at the injection site.

Interactions with other medications

None identified.

Pregnancy and lactation

The product can be administered to beef cows and ewes at any stage of pregnancy or lactation. The product can be administered to sows at any stage of pregnancy or lactation.

Adverse effects

Transitory discomfort has been observed in some cattle following subcutaneous administration. Tissue swellings at the injection site have been observed. These reactions resolve without treatment.

Incompatibility

None known

Overdose and treatment**Cattle**

Single doses of 4.0 mg/kg ivermectin (20 times the recommended dosage) administered subcutaneously, results in ataxia and depression. No antidote has been identified. Symptomatic treatment may be beneficial.

Sheep

Dose levels of up to 4mg/kg given subcutaneously can result in ataxia and depression. Symptomatic treatment may be beneficial.

Pigs

Ivermectin has a recognised wide safety margin and is known to be safe in all ages of swine. It has no adverse effects on fertility in sows or breeding performance of boars.

Clinical signs of ivermectin toxicity in swine include tremors, bilateral mydriasis and recumbency with some biochemical abnormalities including a transient depression of serum iron. Such changes were only observed when ivermectin was administered subcutaneously at a dose of 30 mg/kg (100 times the normal therapeutic dose).

Withdrawal period

Cattle must not be slaughtered for human consumption until 49 days after the last treatment.

This product should not be used in cattle producing milk for human consumption. The product should not be used in non-lactating dairy cows including pregnant heifers within 60 days of calving.

Sheep must not be slaughtered for human consumption until 42 days after the last treatment.

This product should not be used in lactating ewes producing milk for human consumption.

Pigs must not be slaughtered for human consumption until 28 days after the last treatment.

Storage conditions: Store at a temperature not exceeding 30°C. Protect from light

Batch No., Mfg. date, Exp. date: See on the label.

Net Volume: 50 ml & 100 ml vial

NOT FOR HUMAN USE. FOR ANIMAL TREATMENT ONLY



Manufactured by:
INTAS PHARMACEUTICALS LIMITED
C-1/B-53, G.I.D.C ESTATE
NADIAD-387001
Dist : KHEDA
GUJARAT STATE , INDIA

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